

## Exercise 1: Loyalty System on Ethereum

### 1. Fee and Operation Mechanism (Gas Fee)

When users transfer Loyalty Points to each other, they must pay a fee calculated in Gas.

The main purpose of the Gas charging mechanism is for network safety: Gas is a unit of measurement for the computational effort required to execute an order in a Smart Contract. Gas fees are established with the important purpose of preventing infinite loops and spam attacks on the network. By requiring users to pay for each computational operation, the Gas mechanism compensates miners (or validators) and ensures that no one can use network resources indefinitely without limitation.

2. Distinguishing Account Roles The Ethereum system uses two main types of accounts:

| Classification                 | Name (Role in the system)                                    | Characteristics                                                                                                                |
|--------------------------------|--------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------|
| Externally owned account (EOA) | User (Customer)                                              | Private Key is required to sign<br>Receive transactions and control accounts.<br>Used to send Ether and initiate transactions. |
| Contract Account               | The source code that defines the game rules (Smart Contract) | Controlled by source code (Code). No Private Key. Runs automatically when a transaction is received.                           |

The account that is required to have a Private Key to confirm transactions is an Externally Owned Account (EOA), because this is how users control their assets and authorize transactions.

## Exercise 2: Hyperledger Fabric 1. Permissioned Network

Unlike Ethereum, which is a public network, Hyperledger Fabric operates on a Permissioned model.

- If an unknown financial institution wants to participate in this network, they cannot do so freely. All members participating in the network must have their identities authenticated (Known Identities).
- The component in the Fabric architecture responsible for managing identities and issuing digital certificates to members is the Membership Service Provider (MSP).

## **2. Privacy & Channels**

The problem is that Bank C is a competitor, so A and B absolutely do not want C to see their transaction data.

Hyperledger Fabric solution provides Channels feature.

- A and B need to create a private Channel, defined as a private subnet between only these two specific members.
- When A and B execute separate credit agreements, the transaction is recorded only on that Channel's ledger.
- This feature ensures that transactions in Channel A-B are completely invisible to non-Channel members (like Bank C), even though they are all on the same Hyperledger network.

## **3. Cost and Currency (No Cryptocurrency)**

- For the Fabric network of 3 banks, they do not need to issue or buy any virtual currency (Coin) to pay transaction or mining fees, because Fabric does not use Cryptocurrency.
- This feature is very suitable for the business model (B2B) because:

1. Nó giảm thiểu sự phức tạp về mặt quản lý tài chính và tuân thủ pháp lý, vốn là rào cản lớn khi doanh nghiệp sử dụng tiền mã hóa (cryptocurrency).

2. Fabric được thiết kế để phục vụ môi trường B2B (Doanh nghiệp với Doanh nghiệp), nơi mà sự riêng tư, tính linh hoạt của kiến trúc module và khả năng cấp quyền được ưu tiên hơn so với tính công khai của Ethereum (vốn phù hợp hơn với B2C/DApps).